1973A&A....23..411T

erties of the elements in the periodic table, and one of the purposes of the present study is to clarify this relationship for selected representative elements.

II. The Equilibrium Constant and the Thermochemical Data

We assume that the stellar atmosphere is in local thermodynamical equilibrium, and that the chemical equilibrium at each point in a stellar atmosphere can be treated as that of an ideal gas at constant pressure and temperature. We consider a gaseous reaction in which a molecule $X_l\ Y_m\ Z_n$ is formed from its constituent free atoms X, Y and Z:

$$X_l Y_m Z_n = lX + mY + nZ\,. \tag{1}$$

For this reaction, the equilibrium constant K_p can be defined by

$$K_p = \frac{P_X^l P_Y^m P_Z^n}{P_{X_l Y_m Z_n}}, \tag{2}$$

where P_X, P_Y, P_Z, and $P_{X_l Y_m Z_n}$ are the partial pressures of X, Y, Z, and $X_l Y_m Z_n$ respectively. The equilibrium constant, K_p, can be calculated from

$$\ln K_p = l\left[-\frac{F^0-H_0^0}{RT}\right]_X + m\left[-\frac{F^0-H_0^0}{RT}\right]_Y + n\left[-\frac{F^0-H_0^0}{RT}\right]_Z - \left[-\frac{F^0-H_0^0}{RT}\right]_{X_l Y_m Z_n} - \frac{\Delta H_0^0(X_l Y_m Z_n)}{RT}, \tag{3}$$

where $\Delta H_0^0(X_l Y_m Z_n)$ is the heat of atomization of $X_l Y_m Z_n$ and $[-(F^0-H_0^0)/RT]$'s are the free energy functions.

The most important molecular data in calculating the equilibrium constant is the heat of atomization or the dissociation energy. The heats of atomization adopted in the present study are summarized in Table 1. Some of these data were extensively discussed in a monograph by Gaydon (1968) and in compilations by Chao *et al.* (1965 – 1968). For these species, we refer only to one of these reviews, and in general not to the original literature. When the values adopted in these two sources differ, we adopt the one which seems to be based on well-founded experimental data. Sometimes, it was difficult to judge which was the better, both values being based on Birge-Sponer extrapolation which does not give an unique solution, or being estimated by analogy with a similar bonding. In such cases, we use a mean value, and we refer to both sources. For more recent data and for species not covered by these reviews, we refer to the original literature. Moreover, the heats of atomization of some of the important molecules had to be estimated. We comment on these cases below.

For calculating the free-energy functions, it is also necessary to have structural data of the molecular species. Such data are available in several compilations (e.g. Herzberg, 1945, 1967; McBride *et al.*, 1963; Chao *et al.*, 1965 – 1968), or in the literature cited in Table 1. If no such data are available, they are estimated: bond distances are estimated by analogy with similar bonding, vibrational frequencies are calculated by the valence force model with force constants estimated by analogy with similar bonding, and electronic multiplicities are assumed to be the same as those of the similar electronic configuration.

Molecules Composed of H, C, N *and* O. The dissociation energies for these species are discussed in Paper I and we rediscuss only CN. The dissociation energy of CN has long been at issue, and values of 7.5, 7.9 and 8.2 eV have been suggested. In paper I, we adopted the highest value. Gaydon (1968) has reviewed the problem in detail and has shown that the situation is still confusing. He was finally led to favour an intermediate value and recommended $D(\mathrm{C-N}) = 7.75 \pm 0.2$ eV (178 kcal/mol). Recently, Dibeler and Liston (1968) have indicated $D(\mathrm{C-N}) = 7.9$ eV (182.5 kcal/mol). They cited several other experimental results that agree with their own and also remarked that any systematic errors in their analysis will lead to their value being a lower limit to the true value. Accordingly, we use the value of $D(\mathrm{C-N}) = 7.9$ eV (182 kcal/mol) in the present study.

Other species for which the heats of atomization should be revised more than 0.05 eV from the values adopted in Paper I are the following:

NH (88 kcal/mol → 74 kcal/mol),

C_2H (273 kcal/mol → 278 kcal/mol),

and C_2N (329 kcal/mol → 320 kcal/mol).

For the other species composed of H, C, N and O, the dissociation energies adopted in Paper I are essentially unchanged. Equilibrium constants, however, are all recalculated based on new thermochemical data, if available, and on the new definition of the equilibrium constant given by Eq. (2). In paper I, the equilibrium constant was defined for a binary reaction, whilst, in the present study, it is defined for the formation of molecules from gaseous, neutral atoms. The two definitions give identical values only for diatomic molecules.

Hydrides. For most metallic hydrides, dissociation energies were taken from Gaydon (1968). They have mostly been determined by the Birge-Sponer extrapolation and by predissociation.

Carbides. From laboratory work, little is known about gaseous metal monocarbides. Gaydon (1968) has cited an upper limit to $D(\mathrm{Ti-C})$ of 5.75 eV. This value is based on a mass spectrometic study by Chupka *et al.* (1958) who have pointed out that the intensity peak

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1973A&A....23..411T

Table 2. The standard chemical composition

Element	log N	Element	log N	Element	log N
H	12.00	Cl	5.50: [6]	Sc	3.04 [7]
He	11.21:	Cr	5.47 [7]	Sr	2.82 [4]
O	8.77 [1]	P	5.43 [5]	Br	2.68: [6]
C	8.55 [1]	Ni	5.08 [7]	Zr	2.65 [9]
N	7.93 [1]	K	5.05 [5]	Rb	2.63 [10]
Fe	7.62 [2]	Mn	4.88 [7]	La	2.03 [9]
Si	7.55 [3]	F	4.75: [6]	Nd	1.93 [9]
Mg	7.48 [4]	Ti	4.50 [7]	Ba	1.90 [4]
S	7.21 [5]	V	3.92 [7]	Ce	1.78 [9]
Al	6.40 [5]	B	3.6: [8]	I	1.45: [6]
Ca	6.33 [4]	Cu	3.50 [9]	Be	1.1 [11]
Na	6.18 [5]	Y	3.20: [9]	Li	0.68: [12]

1. Lambert (1968). 2. Baschek *et al.* (1970); Garz *et al.* (1970). 3. Lambert and Warner (1968b). 4. Lambert and Warner (1968c). 5. Lambert and Warner (1968a). 6. Suess and Urey (1956). 7. Warner (1968). 8. Lambert *et al.* (1966). 9. Goldberg *et al.* (1960). 10. Lambert and Mallia (1968). 11. Grevesse (1968); Hauge and Engvold (1969). 12. Peach (1968); Wiehr *et al.* (1968).

1934) which made use of equilibrium constants to express the mass balance constraint for all the elements in terms of the partial pressures of free neutral atoms. This method is simple enough to allow a general treatment of the complicated problem mentioned above. The computer program has been written in such a way that any species and any element can be included in the equilibrium by just adding a data card specifying the molecular formula and equilibrium constant. The chemical equilibrium in a complicated system can also be solved by the minimization of the total free-energy of the system (White *et al.*, 1958). It is, however, unlikely that this method has any marked advantage over that based on the classical Russell equations in so far as the chemical equilibrium in stellar atmosphere is concerned.

In the present study, negative ion formation by both atoms and molecules as well as single ionization of atoms are taken into account and solved for simultaneously with the molecular equilibrium. We have, however, investigated only gaseous systems, and the formation of solids and liquids have not been considered.

In the actual computations, we first determine the fictitious pressure of the hydrogen nucleus, $P(\mathrm{H})$, defined by

$$P(\mathrm{H}) = P_{\mathrm{H}} + P_{\mathrm{H}+} + 2P_{\mathrm{H}_2}, \tag{4}$$

where P_{H}, $P_{\mathrm{H}+}$, and P_{H_2} are the partial pressures of neutral hydrogen, ionized hydrogen, and molecular hydrogen, respectively. To calculate $P(\mathrm{H})$ from the above equation, we must calculated P_{H} for a given temperature and gas pressure from

$$P_g = P_{\mathrm{H}} + P_{\mathrm{H}+} + P_{\mathrm{H}_2} + P_{\mathrm{He}} + P_e, \tag{5}$$

where P_{He} is the partial pressure of neutral helium and is equated to the fictitious pressure of the helium nuclei. The electron pressure, P_e, is equated to $P_{\mathrm{H}+}$, since otherwise the electron pressure is so small that it gives no contribution to the gas pressure. Then, the above equation and the ionization and dissociation equations are solved for P_{H}. After $P(\mathrm{H})$ has been obtained by the above procedure, the fictitious pressure of any element, M, can be obtained from

$$P(\mathrm{M}) = a(\mathrm{M})\, P(\mathrm{H}), \tag{6}$$

where $a(\mathrm{M})$ is the abundance of element M relative to hydrogen. These fictitious pressures are necessary in solving the Russell equations. Next, the Russell equations are solved by iteration using the Newton-Raphson method. The convergence is reasonably fast in general, even though hundreds of species are incorporated.

IV. Chemical Equilibrium in Stellar Atmospheres of Normal Chemical Abundance

The chemical equilibria of 36 elements are solved for the standard chemical composition shown in Table 2.

The resulting partial pressures of free neutral atoms for 35 elements are shown in Table 3 for the temperature range between 1008 °K ($\theta = 5.0$) and 6300 °K ($\theta = 0.8$) and for log P_g between 0.0 and 8.0. In this table, the fictitious pressure of hydrogen, $P(\mathrm{H})$, and the electron pressure are also shown. The partial pressure of helium is not shown since it is given simply by $P_{\mathrm{He}} = P(\mathrm{H})\, a(\mathrm{He})$. From the equilibrium partial pressures for neutral atoms, the partial pressure of any molecule can directly be evaluated from equation (2). The necessary equilibrium constants are approximated by

$$\log K_p = a_0 + a_1\theta + a_2\theta^2 + a_3\theta^3 + a_4\theta^4, \tag{7}$$

and the fitting coefficients are given in Table 4.

In Figs. 1a and 1b, we have plotted, for selected elements, $\log n(\mathrm{M})/N(\mathrm{M})$ against $\theta = 5040/T$ for the case of $\log P_g = 3.0$, where $n(\mathrm{M})$ is the number density of the free neutral atom and $N(\mathrm{M})$ is the number density of the nuclei of element M. The decrease of $\log n(\mathrm{M})/N(\mathrm{M})$ is due to the effect of molecular formation at lower temperatures and to ionization at higher temperatures. At lower temperatures, the fraction of neutral atoms is fairly large for Group Ia elements (alkali metals), but it decreases in Group IIa (alkali earth metals) and especially in Group IIIa (Sc, Y, La, and lanthanides) and group IVa (Ti, Zr etc. . .). It then increases again in the transition elements of Group Va (V), VIa (Cr), VIIa (Mn), VIII (Fe, Co, Ni), Ib (Cu), and IIb (Zn). Eventually, the fraction decreases in the non-metallic elements of Group IIIb (B, Al), IVb (C, Si), Vb (N, P), VIb (O, S) and VII b (halogens). It is interesting to note that the elements most strongly locked in molecules are not the ones of the strongest

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1973A&A....23..411T

TABLE 3 LOGARITHMS OF THE EQUILIBRIUM PARTIAL PRESSURE OF GASEOUS NEUTRAL ATOMS (UNIT OF PRESSURE IS DYN/CM**2)

	THETA=	5.0	4.5	4.0	3.5	3.0	2.8	2.6	2.4	2.2	2.0	1.8	1.6	1.4	1.2	1.0	0.8
LOG PG=0.0	ELECTRON	-10.74	-9.58	-8.43	-7.33	-6.58	-6.21	-5.90	-5.71	-5.44	-5.20	-4.72	-4.22	-4.02	-3.69	-2.43	-1.01
	H NUCLEI	0.18	0.18	0.18	0.18	0.16	0.12	0.04	-0.04	-0.06	-0.06	-0.06	-0.06	-0.06	-0.06	-0.07	-0.11
	H	-5.60	-4.45	-3.29	-2.13	-0.99	-0.58	-0.26	-0.10	-0.07	-0.07	-0.06	-0.06	-0.06	-0.06	-0.07	-0.17
	LI	-12.64	-11.67	-11.19	-11.15	-11.17	-11.23	-11.44	-12.05	-12.90	-13.83	-14.55	-15.25	-16.27	-17.19	-17.20	-17.15
	BE	-16.70	-14.16	-11.69	-10.74	-10.74	-10.78	-10.86	-10.94	-10.96	-10.96	-10.96	-10.97	-11.26	-12.66	-13.45	-14.19
	B	-26.26	-23.01	-19.76	-16.61	-14.38	-13.80	-13.29	-12.25	-10.63	-9.08	-8.49	-8.49	-8.97	-10.31	-10.90	-11.43
	C	-29.21	-26.20	-23.19	-20.18	-17.22	-16.11	-15.05	-13.45	-11.26	-9.03	-6.80	-4.61	-3.57	-3.55	-3.85	-4.74
	N	-20.41	-17.94	-15.47	-13.00	-10.54	-9.57	-8.62	-7.66	-6.67	-5.68	-4.73	-4.18	-4.14	-4.13	-4.14	-4.23
	O	-17.06	-14.46	-11.86	-9.25	-6.61	-5.51	-4.40	-3.82	-3.77	-3.73	-3.69	-3.64	-3.32	-3.30	-3.30	-3.39
	F	-20.82	-18.66	-16.51	-14.08	-11.93	-11.16	-10.34	-9.35	-8.24	-7.48	-7.33	-7.32	-7.31	-7.31	-7.32	-7.36
	NA	-5.65	-5.64	-5.64	-5.64	-5.69	-5.82	-6.25	-7.09	-7.94	-8.83	-9.50	-10.15	-11.12	-11.99	-11.95	-11.85
	MG	-4.34	-4.34	-4.34	-4.34	-4.36	-4.40	-4.48	-4.56	-4.58	-4.60	-4.76	-5.50	-6.91	-8.28	-8.75	-9.15
	AL	-9.06	-7.68	-6.34	-5.52	-5.45	-5.48	-5.57	-5.66	-5.84	-6.51	-7.29	-8.10	-9.24	-10.28	-10.41	-10.47
	SI	-16.30	-14.71	-13.11	-11.52	-9.98	-9.44	-8.95	-7.92	-6.31	-4.89	-4.54	-4.65	-5.69	-7.13	-7.70	-8.19
	P	-14.57	-13.30	-11.91	-10.02	-7.78	-7.43	-7.31	-6.90	-6.65	-6.64	-6.63	-6.63	-6.65	-7.14	-8.03	-8.98
	S	-8.91	-7.31	-5.85	-4.89	-4.68	-4.69	-4.76	-4.83	-4.85	-4.85	-4.85	-4.85	-4.86	-5.06	-5.72	-6.60
	CL	-12.20	-10.81	-9.65	-8.45	-7.34	-6.96	-6.69	-6.59	-6.57	-6.57	-6.56	-6.56	-6.56	-6.57	-6.59	-6.96
	K	-6.82	-6.78	-6.78	-6.91	-7.97	-8.56	-9.27	-10.11	-10.83	-11.56	-12.07	-12.56	-13.37	-14.08	-13.88	-13.62
	CA	-5.90	-5.58	-5.51	-5.49	-5.51	-5.55	-5.64	-5.82	-6.36	-7.35	-8.20	-9.04	-10.21	-11.27	-11.43	-11.52
	SC	-22.36	-20.20	-18.05	-15.93	-13.90	-13.20	-12.57	-10.95	-9.33	-9.74	-10.60	-11.52	-12.77	-13.92	-14.17	-14.34
	TI	-15.03	-13.94	-12.86	-11.79	-10.77	-10.43	-10.14	-9.31	-8.06	-7.88	-8.62	-9.57	-10.87	-12.08	-12.38	-12.61
	V	-12.34	-11.13	-10.12	-9.32	-8.70	-8.55	-8.48	-8.23	-8.17	-8.44	-9.17	-10.10	-11.39	-12.58	-12.86	-13.08
	CR	-6.36	-6.36	-6.35	-6.35	-6.38	-6.41	-6.49	-6.57	-6.61	-6.86	-7.57	-8.50	-9.79	-10.98	-11.27	-11.49
	MN	-6.94	-6.94	-6.94	-6.94	-6.96	-7.00	-7.08	-7.16	-7.18	-7.21	-7.42	-8.19	-9.58	-10.90	-11.33	-11.68
	FE	-4.20	-4.20	-4.20	-4.20	-4.22	-4.26	-4.34	-4.42	-4.44	-4.45	-4.49	-4.90	-6.24	-7.65	-8.16	-8.60
	NI	-6.74	-6.74	-6.74	-6.74	-6.76	-6.80	-6.88	-6.96	-6.98	-6.99	-7.04	-7.43	-8.71	-10.07	-10.54	-10.93
	CU	-8.33	-8.32	-8.32	-8.32	-8.34	-8.38	-8.46	-8.54	-8.56	-8.57	-8.60	-8.93	-10.19	-11.57	-12.05	-12.46
	BR	-12.35	-10.89	-9.86	-9.34	-9.20	-9.22	-9.29	-9.36	-9.38	-9.38	-9.38	-9.38	-9.38	-9.40	-9.61	-10.50
	RB	-9.29	-9.21	-9.23	-9.56	-10.85	-11.43	-12.11	-12.92	-13.60	-14.30	-14.78	-15.24	-16.02	-16.69	-16.46	-16.17
	SR	-11.01	-9.28	-9.01	-9.00	-9.03	-9.08	-9.25	-9.82	-10.70	-11.69	-12.46	-13.23	-14.31	-15.29	-15.36	-15.37
	Y	-25.43	-22.96	-20.51	-18.09	-15.74	-14.91	-14.16	-12.34	-9.78	-9.10	-9.63	-10.45	-11.66	-12.78	-12.99	-13.14
	ZR	-23.33	-21.21	-19.14	-17.18	-15.40	-14.80	-14.27	-13.16	-11.51	-10.08	-10.44	-11.39	-12.69	-13.90	-14.21	-14.44
	I	-10.52	-10.39	-10.38	-10.37	-10.39	-10.43	-10.51	-10.59	-10.61	-10.61	-10.61	-10.61	-10.62	-11.03	-11.87	-12.81
	BA	-14.61	-12.33	-10.74	-10.10	-10.03	-10.25	-10.84	-11.78	-12.66	-13.57	-14.25	-14.92	-15.90	-16.78	-16.76	-16.67
	LA	-30.82	-27.93	-25.04	-22.16	-19.35	-18.34	-17.40	-15.40	-12.71	-12.67	-13.42	-14.17	-15.23	-16.20	-16.25	-16.24
	CE	-31.12	-28.23	-25.34	-22.46	-19.65	-18.64	-17.71	-15.71	-13.03	-11.27	-11.37	-12.34	-13.65	-14.88	-15.19	-15.44
	ND	-28.24	-25.57	-22.90	-20.25	-17.68	-16.77	-15.93	-14.01	-11.14	-11.18	-12.84	-12.92	-14.12	-15.22	-15.82	-15.55
LOG PG=1.0	ELECTRON	-10.26	-9.08	-7.93	-6.78	-5.84	-5.55	-5.17	-4.86	-4.64	-4.37	-4.05	-3.49	-3.09	-2.93	-1.91	-0.48
	H NUCLEI	1.18	1.18	1.18	1.18	1.17	1.16	1.12	1.04	0.96	0.94	0.93	0.93	0.93	0.93	0.93	0.92
	H	-5.10	-3.95	-2.79	-1.63	-0.48	-0.02	0.40	0.73	0.89	0.93	0.93	0.93	0.93	0.93	0.93	0.90
	LI	-11.90	-11.04	-10.28	-10.15	-10.15	-10.17	-10.23	-10.47	-11.14	-12.01	-12.87	-13.52	-14.34	-15.43	-15.68	-15.58
	BE	-16.70	-14.16	-11.64	-9.85	-9.73	-9.76	-9.78	-9.86	-9.94	-9.96	-9.96	-9.97	-10.01	-10.94	-11.93	-12.62
	B	-25.76	-22.51	-19.26	-16.05	-13.43	-12.72	-12.14	-11.60	-10.54	-8.94	-7.68	-7.47	-7.56	-8.58	-9.38	-9.86
	C	-28.21	-25.20	-22.19	-19.18	-16.19	-15.01	-13.89	-12.80	-11.18	-8.99	-6.78	-4.53	-2.80	-2.52	-2.65	-3.24
	N	-19.91	-17.44	-14.97	-12.50	-10.03	-9.05	-8.07	-7.12	-6.16	-5.17	-4.19	-3.36	-3.14	-3.13	-3.14	-3.16
	O	-17.06	-14.46	-11.86	-9.25	-6.63	-5.57	-4.48	-3.38	-2.82	-2.77	-2.71	-2.69	-2.44	-2.30	-2.30	-2.32
	F	-20.32	-18.16	-16.04	-13.77	-11.46	-10.68	-9.91	-9.10	-8.13	-7.05	-6.42	-6.32	-6.32	-6.31	-6.32	-6.33
	NA	-4.65	-4.64	-4.64	-4.64	-4.65	-4.69	-4.83	-5.29	-6.14	-7.00	-7.83	-8.42	-9.19	-10.23	-10.43	-10.28
	MG	-3.34	-3.34	-3.34	-3.34	-3.35	-3.36	-3.40	-3.48	-3.56	-3.58	-3.63	-3.96	-5.00	-6.53	-7.23	-7.58
	AL	-8.56	-7.17	-5.81	-4.69	-4.44	-4.45	-4.48	-4.56	-4.67	-4.94	-5.65	-6.38	-7.31	-8.52	-8.89	-8.91
	SI	-15.30	-13.71	-12.12	-10.53	-8.95	-8.34	-7.79	-7.27	-6.23	-4.65	-3.63	-3.55	-3.94	-5.38	-6.17	-6.63
	P	-13.57	-12.32	-11.04	-9.54	-7.37	-6.58	-6.21	-6.08	-5.77	-5.64	-5.64	-5.63	-5.64	-5.78	-6.54	-7.42
	S	-8.90	-7.30	-5.72	-4.38	-3.76	-3.70	-3.70	-3.76	-3.83	-3.85	-3.85	-3.85	-3.86	-3.90	-4.31	-5.06
	CL	-11.93	-10.35	-9.15	-7.97	-6.81	-6.38	-5.99	-5.71	-5.59	-5.57	-5.57	-5.56	-5.56	-5.56	-5.57	-5.71
	K	-5.86	-5.79	-5.78	-5.81	-6.33	-6.88	-7.46	-8.18	-9.00	-9.73	-10.39	-10.83	-11.44	-12.32	-12.36	-12.05
	CA	-5.27	-4.73	-4.54	-4.50	-4.50	-4.51	-4.55	-4.65	-4.89	-5.57	-6.53	-7.32	-8.28	-9.51	-9.91	-9.96
	SC	-21.36	-19.20	-17.06	-14.93	-12.84	-12.05	-11.34	-10.67	-9.16	-8.29	-8.97	-9.80	-10.85	-12.16	-12.64	-12.78
	TI	-14.03	-12.94	-11.86	-10.79	-9.73	-9.33	-8.98	-8.66	-7.82	-6.82	-7.07	-7.86	-8.94	-10.32	-10.85	-11.04
	V	-11.34	-10.14	-9.12	-8.32	-7.66	-7.47	-7.34	-7.29	-7.17	-7.20	-7.63	-8.40	-9.46	-10.82	-11.34	-11.51
	CR	-5.39	-5.37	-5.36	-5.36	-5.36	-5.37	-5.41	-5.49	-5.57	-5.64	-6.04	-6.79	-7.86	-9.22	-9.75	-9.93
	MN	-5.94	-5.94	-5.94	-5.94	-5.95	-5.96	-6.00	-6.08	-6.16	-6.18	-6.25	-6.62	-7.66	-9.15	-9.80	-10.12
	FE	-3.20	-3.20	-3.20	-3.20	-3.21	-3.22	-3.26	-3.34	-3.42	-3.44	-3.46	-3.58	-4.36	-5.89	-6.64	-7.04
	NI	-5.74	-5.74	-5.74	-5.74	-5.75	-5.76	-5.80	-5.88	-5.96	-5.98	-6.00	-6.11	-6.84	-8.32	-9.01	-9.37
	CU	-7.34	-7.33	-7.32	-7.32	-7.33	-7.34	-7.38	-7.46	-7.54	-7.56	-7.57	-7.66	-8.33	-9.81	-10.52	-10.89
	BR	-12.04	-10.62	-9.37	-8.61	-8.26	-8.21	-8.22	-8.29	-8.36	-8.38	-8.38	-8.38	-8.38	-8.39	-8.47	-9.02
	RB	-8.35	-8.23	-8.21	-8.33	-9.15	-9.74	-10.30	-10.99	-11.78	-12.48	-13.11	-13.51	-14.09	-14.93	-14.94	-14.60
	SR	-10.53	-8.90	-8.04	-8.01	-8.01	-8.03	-8.08	-8.29	-8.95	-9.87	-10.79	-11.50	-12.38	-13.53	-13.84	-13.80
	Y	-24.42	-21.96	-19.51	-17.08	-14.69	-13.76	-12.92	-12.11	-10.29	-8.26	-8.18	-8.77	-9.74	-11.02	-11.47	-11.57
	ZR	-22.33	-20.21	-18.14	-16.18	-14.36	-13.69	-13.09	-12.54	-11.44	-9.83	-8.97	-9.68	-10.77	-12.14	-12.68	-12.88
	I	-9.96	-9.45	-9.39	-9.38	-9.38	-9.39	-9.43	-9.51	-9.59	-9.61	-9.61	-9.61	-9.62	-9.72	-10.40	-11.25
	BA	-14.14	-12.19	-10.17	-9.32	-9.00	-9.04	-9.28	-9.91	-10.85	-11.74	-12.58	-13.19	-13.97	-15.03	-15.24	-15.11
	LA	-29.82	-26.93	-24.04	-21.16	-18.30	-17.19	-16.16	-15.17	-13.18	-11.12	-11.75	-12.44	-13.30	-14.44	-14.73	-14.68
	CE	-30.12	-27.23	-24.34	-21.46	-18.60	-17.49	-16.46	-15.48	-13.50	-11.19	-9.96	-10.63	-11.73	-13.12	-13.67	-13.88
	ND	-27.84	-24.57	-21.91	-19.25	-16.63	-15.62	-14.69	-13.80	-11.86	-9.66	-10.39	-11.20	-12.19	-13.47	-13.90	-13.98
LOG PG=2.0	ELECTRON	-9.79	-8.60	-7.43	-6.27	-5.18	-4.84	-4.51	-4.12	-3.81	-3.57	-3.29	-2.84	-2.28	-2.02	-1.36	0.04
	H NUCLEI	2.18	2.18	2.18	2.18	2.18	2.17	2.16	2.12	2.05	1.96	1.94	1.93	1.93	1.93	1.93	1.93
	H	-4.60	-3.45	-2.29	-1.13	0.03	0.49	0.95	1.37	1.71	1.89	1.93	1.93	1.93	1.93	1.93	1.92
	LI	-11.14	-10.41	-9.48	-9.17	-9.15	-9.15	-9.17	-9.24	-9.51	-10.23	-11.11	-11.87	-12.53	-13.51	-14.13	-14.06
	BE	-16.70	-14.16	-11.63	-9.30	-8.74	-8.73	-8.74	-8.78	-8.85	-8.93	-8.96	-8.96	-8.97	-9.27	-10.40	-11.10
	B	-25.26	-22.00	-18.76	-15.53	-12.59	-11.75	-11.07	-10.47	-9.91	-8.84	-7.30	-6.52	-6.48	-6.85	-7.85	-8.34
	C	-27.21	-24.20	-21.19	-18.18	-15.18	-13.98	-12.80	-11.67	-10.55	-8.91	-6.74	-4.52	-2.38	-1.54	-1.56	-1.87
	N	-19.41	-16.94	-14.47	-12.00	-9.53	-8.54	-7.56	-6.58	-5.62	-4.66	-3.68	-2.73	-2.18	-2.14	-2.14	-2.14
	O	-17.06	-14.46	-11.86	-9.25	-6.64	-5.58	-4.52	-3.44	-2.37	-1.83	-1.75	-1.70	-1.61	-1.31	-1.30	-1.30
	F	-19.82	-17.66	-15.55	-13.40	-11.02	-10.20	-9.43	-8.66	-7.86	-6.91	-5.88	-5.38	-5.32	-5.32	-5.31	-5.32
	NA	-3.66	-3.65	-3.64	-3.64	-3.64	-3.65	-3.69	-3.85	-4.34	-5.19	-6.06	-6.77	-7.39	-8.31	-8.89	-8.76
	MG	-2.35	-2.34	-2.34	-2.34	-2.34	-2.35	-2.36	-2.40	-2.47	-2.55	-2.59	-2.70	-3.28	-4.61	-5.68	-6.06
	AL	-8.05	-6.67	-5.30	-4.02	-3.47	-3.44	-3.44	-3.48	-3.56	-3.69	-4.06	-4.75	-5.51	-6.60	-7.34	-7.38
	SI	-14.30	-12.71	-11.13	-9.54	-7.94	-7.31	-6.71	-6.14	-5.60	-4.53	-3.07	-2.55	-2.61	-3.50	-4.63	-5.11
	P	-12.57	-11.32	-10.07	-8.75	-7.03	-6.15	-5.39	-5.02	-4.89	-4.69	-4.64	-4.63	-4.63	-4.65	-5.12	-5.91
	S	-8.90	-7.29	-5.69	-4.16	-3.05	-2.84	-2.74	-2.71	-2.76	-2.83	-2.85	-2.85	-2.85	-2.86	-3.04	-3.59
	CL	-11.70	-9.96	-8.66	-7.49	-6.31	-5.85	-5.42	-5.02	-4.72	-4.59	-4.57	-4.57	-4.56	-4.56	-4.57	-4.62
	K	-4.91	-4.82	-4.78	-4.78	-4.97	-5.29	-5.80	-6.37	-7.09	-7.90	-8.63	-9.18	-9.64	-10.40	-10.82	-10.53
	CA	-4.71	-4.02	-3.63	-3.52	-3.50	-3.50	-3.51	-3.55	-3.65	-3.99	-4.80	-5.67	-6.48	-7.60	-8.37	-8.43
	SC	-20.35	-18.19	-16.05	-13.93	-11.83	-11.00	-10.21	-9.49	-8.81	-7.58	-7.41	-8.17	-9.05	-10.25	-11.10	-11.25
	TI	-13.03	-11.94	-10.86	-9.79	-8.72	-8.30	-7.89	-7.52	-7.17	-6.36	-5.76	-6.28	-7.15	-8.40	-9.31	-9.52
	V	-10.35	-9.14	-8.12	-7.32	-6.66	-6.44	-6.28	-6.19	-6.17	-6.15	-6.27	-6.82	-7.67	-8.90	-9.80	-9.99
	CR	-4.47	-4.41	-4.38	-4.36	-4.36	-4.36	-4.38	-4.41	-4.48	-4.57	-4.71	-5.23	-6.07	-7.31	-8.20	-8.40
	MN	-4.94	-4.94	-4.94	-4.94	-4.94	-4.95	-4.96	-5.00	-5.07	-5.15	-5.19	-5.32	-5.93	-7.23	-8.26	-8.59
	FE	-2.20	-2.20	-2.20	-2.20	-2.20	-2.21	-2.22	-2.26	-2.33	-2.41	-2.44	-2.48	-2.78	-3.99	-5.10	-5.51
	NI	-4.74	-4.74	-4.74	-4.74	-4.74	-4.75	-4.76	-4.80	-4.87	-4.95	-4.98	-5.02	-5.28	-6.41	-7.47	-7.84
	CU	-6.38	-6.35	-6.33	-6.33	-6.33	-6.33	-6.34	-6.38	-6.45	-6.53	-6.56	-6.59	-6.81	-7.91	-8.98	-9.37
	BR	-11.74	-10.32	-8.99	-8.02	-7.42	-7.30	-7.24	-7.23	-7.29	-7.36	-7.38	-7.38	-7.38	-7.39	-7.41	-7.68
	RB	-7.44	-7.29	-7.21	-7.24	-7.63	-8.06	-8.62	-9.17	-9.86	-10.64	-11.34	-11.86	-12.28	-13.02	-13.40	-13.08
	SR	-10.00	-8.62	-7.17	-7.02	-7.01	-7.01	-7.03	-7.09	-7.35	-8.09	-9.03	-9.84	-10.57	-11.61	-12.29	-12.28
	Y	-23.42	-20.96	-18.51	-16.08	-13.67	-12.72	-11.79	-10.93	-10.08	-8.33	-7.05	-7.28	-7.96	-9.11	-9.93	-10.05
	ZR	-21.33	-19.21	-17.14	-15.18	-13.35	-12.66	-12.00	-11.40	-10.83	-9.73	-8.21	-8.11	-8.97	-10.23	-11.14	-11.35
	I	-9.75	-8.68	-8.43	-8.39	-8.38	-8.38	-8.39	-8.43	-8.50	-8.58	-8.61	-8.61	-8.61	-8.63	-9.00	-9.75
	BA	-13.60	-11.96	-9.76	-8.61	-8.08	-8.01	-8.05	-8.31	-8.98	-9.91	-10.81	-11.53	-12.17	-13.11	-13.70	-13.58
	LA	-28.81	-25.92	-23.04	-20.16	-17.28	-16.14	-15.03	-13.99	-12.96	-11.06	-10.03	-10.79	-11.50	-12.52	-13.19	-13.15
	CE	-29.12	-26.23	-23.34	-20.46	-17.58	-16.45	-15.34	-14.30	-13.27	-11.39	-9.45	-9.05	-9.93	-11.20	-12.13	-12.35
	ND	-26.24	-23.57	-20.90	-18.25	-15.62	-14.57	-13.56	-12.62	-11.67	-9.73	-8.74	-9.56	-10.39	-11.55	-12.35	-12.46

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TABLE 4 FITTING COEFFICIENTS TO LOK KP

MOLECULE	A(0)	A(1)	A(2)	A(3)	A(4)	MOLECULE	A(0)	A(1)	A(2)	A(3)	A(4)
HH	1.2739E 01	-5.1172E 00	1.2572E-01	-1.4149E-02	6.3021E-04	MGCL	1.1318E 01	-4.2224E 00	2.1137E-01	-3.0174E-02	1.5480E-03
LIH	1.1782E 01	-3.4443E 00	3.2741E-01	-4.8456E-02	2.5336E-03	MGCL2	2.4666E 01	-8.9557E 00	2.6264E-01	-3.9593E-02	2.0922E-03
LIO	1.2362E 01	-4.5737E 00	3.4069E-01	-5.0059E-02	2.6013E-03	CACL	1.2314E 01	-5.1814E 00	5.6532E-01	-8.2868E-02	4.2822E-03
LIF	1.2449E 01	-6.8550E 00	2.8577E-01	-4.1671E-02	2.1577E-03	CACL2	2.4611E 01	-1.0617E 01	4.7397E-01	-7.0982E-02	3.7140E-03
LICL	1.2014E 01	-5.7321E 00	2.5558E-01	-3.6937E-02	1.9054E-03	ALCL	1.1976E 01	-5.2228E 00	-1.0263E-02	3.9344E-03	-2.6236E-04
LIBR	1.2069E 01	-5.2789E 00	2.7930E-01	-4.0121E-02	2.0533E-03	ALOCL	2.5274E 01	-1.1369E 01	1.8101E-01	-2.6934E-02	1.4130E-03
LII	1.1937E 01	-4.5688E 00	2.7869E-01	-4.0036E-02	2.0486E-03	SIH3CL	4.8899E 01	-1.5218E 01	-4.0856E-03	3.1518E-03	-1.7176E-04
LIOH	2.5122E 01	-1.0625E 01	5.0357E-01	-7.2141E-02	3.7183E-03	KH	1.1417E 01	-2.8459E 00	2.8707E-01	-4.0964E-02	2.0897E-03
BEH	1.1416E 01	-3.1385E 00	2.3509E-01	-3.2702E-02	1.6565E-03	KO	1.1940E 01	-4.0743E 00	4.4901E-01	-6.5741E-02	3.3940E-03
BEH2	2.4160E 01	-7.3693E 00	1.4740E-01	-2.0183E-02	1.0453E-03	KOH	2.4798E 01	-9.7458E 00	4.7803E-01	-6.7612E-02	3.4511E-03
BEO	1.2695E 01	-5.4813E 00	2.5206E-01	-3.5618E-02	1.8152E-03	CAH	1.1340E 01	-3.0144E 00	4.2349E-01	-6.1467E-02	3.1639E-03
BEF	1.2093E 01	-7.1319E 00	2.2863E-01	-3.2040E-02	1.6248E-03	CAO	1.2260E 01	-6.0525E 00	5.8284E-01	-8.5805E-02	4.4425E-03
BECL	1.1749E 01	-4.8045E 00	2.3793E-01	-3.3553E-02	1.7088E-03	CAS	1.2309E 01	-4.9728E 00	6.0546E-01	-8.8967E-02	4.5993E-03
BECL2	2.4864E 01	-1.0132E 01	1.4191E-01	-2.0619E-02	1.0798E-03	CAOH	2.4611E 01	-1.0910E 01	6.0803E-01	-8.7197E-02	4.4736E-03
BEOH	2.4521E 01	-1.0424E 01	2.8415E-01	-3.8203E-02	1.9178E-03	CA(OH)2	4.9549E 01	-1.9354E 01	4.5536E-01	-6.5841E-02	3.4252E-03
BE(OH)2	4.9720E 01	-2.0514E 01	1.1864E-01	-1.5388E-02	8.0412E-04	SCO	1.3747E 01	-8.6420E 00	4.8072E-01	-6.9670E-02	3.5747E-03
BH	1.2399E 01	-3.9819E 00	1.3174E-01	-1.6723E-02	8.1003E-04	SCO2	2.6909E 01	-1.5824E 01	3.9999E-01	-5.9363E-02	3.0875E-03
BH2	2.4368E 01	-8.5831E 00	4.5053E-02	-3.9934E-03	1.7376E-04	SCS	1.3500E 01	-6.4640E 00	5.0458E-01	-7.3117E-02	3.7488E-03
BH3	3.7484E 01	-1.2113E 01	6.3985E-02	-3.6538E-03	1.0482E-04	TIC2	2.7018E 01	-1.3534E 01	4.5875E-01	-6.6158E-02	3.3834E-03
BO	1.3111E 01	-8.8640E 00	1.2714E-01	-1.6313E-02	7.9408E-04	TIC4	5.3496E 01	-2.6027E 01	3.3495E-01	-5.0249E-02	2.6337E-03
BO2	2.5766E 01	-1.4065E 01	3.4383E-02	-4.4398E-03	2.2852E-04	TIN	1.4033E 01	-6.4552E 00	5.3710E-01	-7.6826E-02	3.9056E-03
BS	1.2822E 01	-5.8147E 00	1.7034E-01	-2.3020E-02	1.1454E-03	TIO	1.3398E 01	-8.5956E 00	4.0973E-01	-5.7937E-02	2.9287E-03
BF	1.3080E 01	-8.2826E 00	1.2575E-01	-1.6313E-02	7.9645E-04	TIO2	2.7901E 01	-1.4031E 01	4.2156E-01	-6.1271E-02	3.1476E-03
BCL	1.2634E 01	-6.0398E 00	1.2536E-01	-1.6225E-02	7.8846E-04	TIS	1.3316E 01	-6.2216E 00	4.5829E-01	-6.4903E-02	3.2788E-03
HBO	2.5377E 01	-1.1824E 01	5.0180E-02	-5.3856E-03	2.5485E-04	TIF	1.2575E 01	-7.4975E 00	3.8424E-01	-5.6623E-02	2.9179E-03
HBO2	3.8934E 01	-1.9624E 01	9.8403E-02	-1.0894E-02	5.1783E-04	TIF2	2.5682E 01	-1.4600E 01	1.9921E-01	-2.9451E-02	1.5353E-03
C2	1.2804E 01	-6.5178E 00	9.7719E-02	-1.2739E-02	6.2603E-04	TICL	1.2305E 01	-5.6564E 00	3.8665E-01	-5.6760E-02	2.9204E-03
C3	2.5230E 01	-1.4445E 01	1.2547E-01	-1.7390E-02	8.8594E-04	TICL2	2.5157E 01	-1.0936E 01	1.8766E-01	-2.6875E-02	1.3754E-03
CH	1.2135E 01	-4.0760E 00	1.2768E-01	-1.5473E-02	7.2661E-04	TIOF	2.8068E 01	-1.4706E 01	5.9373E-01	-8.2933E-02	4.1629E-03
CH2	2.5052E 01	-8.4195E 00	8.2200E-02	-8.4261E-03	3.8461E-04	TIOCL	2.6387E 01	-1.2475E 01	4.0173E-01	-5.8149E-02	2.9758E-03
CH3	3.7198E 01	-1.3462E 01	8.7214E-02	-7.1230E-03	2.7670E-04	VC2	2.7279E 01	-1.3337E 01	3.6999E-01	-5.2661E-02	2.6894E-03
CH4	5.0777E 01	-1.7911E 01	1.4135E-02	4.4553E-03	-3.0176E-04	VC4	5.3793E 01	-2.5670E 01	2.7307E-01	-4.1437E-02	2.2108E-03
C2H	2.5063E 01	-1.2291E 01	-1.9036E-02	4.4498E-03	-2.3073E-04	VO	1.3811E 01	-7.7520E 00	3.7056E-01	-5.1467E-02	2.5861E-03
C2H2	3.8184E 01	-1.7365E 01	2.1512E-02	-8.8961E-05	-2.8720E-05	VO2	2.7754E 01	-1.4040E 01	3.3613E-01	-4.8215E-02	2.4780E-03
C2H4	6.3350E 01	-2.3854E 01	-2.9033E-02	9.2219E-03	-4.9573E-04	CRH	1.2297E 01	-4.4505E 00	5.0079E-01	-7.1306E-02	3.6224E-03
C3H	4.0791E 01	-2.1762E 01	9.3377E-01	-1.3863E-01	7.4549E-03	CRC2	2.6747E 01	-1.2469E 01	5.5673E-01	-8.1535E-02	4.2067E-03
C2-	2.3845E 01	-1.1825E 01	6.7867E-01	-8.8402E-02	4.2770E-03	CRO	1.3513E 01	-6.1702E 00	5.8769E-01	-8.4809E-02	4.3322E-03
CH-	2.2955E 01	-7.6855E 00	6.5284E-01	-8.3850E-02	4.0331E-03	MNH	1.1736E 01	-3.6834E 00	4.1609E-01	-6.0528E-02	3.1247E-03
N2	1.3590E 01	-1.0585E 01	2.2067E-01	-2.9997E-02	1.4993E-03	MNO	1.3246E 01	-5.4769E 00	4.3125E-01	-6.3093E-02	3.2606E-03
NH	1.2033E 01	-3.8435E 00	1.3629E-01	-1.6643E-02	7.8691E-04	MNS	1.3151E 01	-4.3821E 00	4.5737E-01	-6.6849E-02	3.4521E-03
NH2	2.4603E 01	-8.6300E 00	2.0048E-01	-2.4124E-02	1.1484E-03	FEO	1.2984E 01	-5.3318E 00	3.1745E-01	-4.4565E-02	2.2524E-03
NH3	3.7554E 01	-1.3059E 01	1.2910E-01	-1.2338E-02	5.3429E-04	FES	1.3855E 01	-4.6198E 00	4.1556E-01	-5.9026E-02	2.9960E-03
CN	1.2805E 01	-8.2793E 00	6.4162E-02	-7.3627E-03	3.4666E-04	FEF	1.3057E 01	-5.8254E 00	3.7447E-01	-5.3030E-02	2.6903E-03
HCN	2.5635E 01	-1.3833E 01	1.3827E-01	-1.8122E-02	9.1645E-04	FEF2	2.4970E 01	-1.0609E 01	2.3078E-01	-3.5235E-02	1.8581E-03
C2N	2.5466E 01	-1.4450E 01	1.5941E-01	-2.3830E-02	1.2585E-03	FECL	1.2148E 01	-4.0158E 00	3.4448E-01	-4.9249E-02	2.5098E-03
C2N2	4.1357E 01	-2.2172E 01	1.6700E-01	-2.3499E-02	1.2586E-03	FECL2	2.4485E 01	-9.0331E 00	2.3953E-01	-3.5825E-02	1.8709E-03
NO	1.2831E 01	-7.1964E 00	1.7349E-01	-2.3065E-02	1.1380E-03	FE(OH)2	4.9408E 01	-1.8136E 01	2.0682E-01	-3.0943E-02	1.6499E-03
HNO	2.5258E 01	-9.5542E 00	1.9462E-01	-2.4812E-02	1.2129E-03	NIH	1.2520E 01	-3.4331E 00	1.9630E-01	-2.5377E-02	1.2412E-03
CN-	2.4123E 01	-1.4422E 01	7.0417E-01	-9.2137E-02	4.4771E-03	NIO	1.3013E 01	-5.0171E 00	2.1076E-01	-2.8251E-02	1.4000E-03
O2	1.3228E 01	-5.5181E 00	6.9935E-02	-8.1511E-03	3.7970E-04	CUH	1.2152E 01	-3.9190E 00	3.0976E-01	-4.3389E-02	2.1933E-03
OH	1.2371E 01	-5.0578E 00	1.3822E-01	-1.6547E-02	7.7224E-04	CUO	1.2176E 01	-4.2827E 00	2.0485E-01	-2.8217E-02	1.4075E-03
H2O	2.5420E 01	-1.0522E 01	1.6939E-01	-1.8368E-02	8.1730E-04	CUS	1.2511E 01	-4.2636E 00	3.7750E-01	-5.4068E-02	2.7545E-03
CO	1.3820E 01	-1.1795E 01	1.7217E-01	-2.2888E-02	1.1349E-03	HBR	1.2301E 01	-4.3046E 00	1.1075E-01	-1.3231E-02	6.1881E-04
CO2	2.7478E 01	-1.7098E 01	9.5012E-02	-1.2579E-02	6.4058E-04	NABR	1.1963E 01	-4.5744E 00	3.1557E-01	-4.5930E-02	2.3683E-03
C2O	2.5724E 01	-1.4651E 01	4.6094E-02	-7.1122E-03	4.0529E-04	KBR	1.2088E 01	-5.1959E 00	4.2982E-01	-6.2876E-02	3.2438E-03
HCO	2.5363E 01	-1.3213E 01	1.8451E-01	-2.2973E-02	1.1114E-03	RBBR	1.1294E 01	-4.3923E 00	1.0285E-01	-1.3374E-02	6.4687E-04
HCHO	3.8383E 01	-1.6397E 01	8.3773E-02	-7.4410E-03	3.1337E-04	MGBR	1.1438E 01	-4.0782E 00	2.7446E-01	-3.9608E-02	2.0331E-03
OH-	2.3621E 01	-8.8212E 00	6.3799E-01	-8.0888E-02	3.8591E-03	MGBR2	2.4153E 01	-7.5523E 00	1.7436E-01	-2.6626E-02	1.4081E-03
HF	1.2444E 01	-6.0199E 00	-5.5059E-02	1.3363E-02	-8.1880E-04	CABR	1.2083E 01	-5.3048E 00	4.7701E-01	-6.9871E-02	3.6055E-03
NAF	1.2187E 01	-5.8118E 00	2.7848E-01	-4.0864E-02	2.1223E-03	CABR2	2.3566E 01	-9.5378E 00	4.5884E-01	-6.9266E-02	3.6274E-03
KF	1.2448E 01	-6.3832E 00	4.2974E-01	-6.2720E-02	3.2339E-03	RBH	1.0602E 01	-1.7516E 00	-4.4476E-02	9.2641E-03	-5.4739E-04
MGF	1.1727E 01	-5.4169E 00	2.4672E-01	-3.5752E-02	1.8465E-03	RBF	1.1654E 01	-5.6299E 00	1.0078E-01	-1.2876E-02	6.1709E-04
MGF2	2.5587E 01	-1.1733E 01	3.2973E-01	-5.0152E-02	2.6559E-03	RBCL	1.1366E 01	-4.8565E 00	1.0116E-01	-1.2629E-02	5.9743E-04
CAF	1.2357E 01	-7.1372E 00	5.6024E-01	-8.2266E-02	4.2527E-03	SRH	1.1827E 01	-3.6194E 00	6.7189E-01	-9.8927E-02	5.1226E-03
CAF2	2.4485E 01	-1.3238E 01	5.5677E-01	-8.1420E-02	4.1998E-03	SRO	1.2458E 01	-6.2098E 00	6.9168E-01	-1.0216E-01	5.2955E-03
ALF	1.2756E 01	-7.4947E 00	1.7507E-01	-2.4848E-02	1.2738E-03	SRS	1.2520E 01	-5.2793E 00	7.1405E-01	-1.0527E-01	5.4491E-03
ALOF	2.5830E 01	-1.3590E 01	2.5776E-01	-3.9441E-02	2.1030E-03	SRF	1.2318E 01	-7.2384E 00	5.6010E-01	-8.1994E-02	4.2281E-03
SIH3F	4.9315E 01	-1.7282E 01	5.7304E-03	1.2804E-03	-5.8634E-05	SRF2	2.4400E 01	-1.3345E 01	6.6768E-01	-9.8245E-02	5.0839E-03
NAH	1.1415E 01	-2.7508E 00	1.9646E-01	-2.7383E-02	1.3844E-03	SRCL	1.2231E 01	-5.7706E 00	6.7316E-01	-9.9034E-02	5.1241E-03
NAO	1.1799E 01	-3.8242E 00	3.3266E-01	-4.8423E-02	2.4970E-03	SRCL2	2.3875E 01	-1.1348E 01	5.7865E-01	-8.6621E-02	4.5257E-03
NAOH	2.4604E 01	-9.4104E 00	3.6408E-01	-5.0732E-02	2.5799E-03	SROH	2.4863E 01	-1.1153E 01	7.1421E-01	-1.0306E-01	5.2983E-03
MGH	1.1285E 01	-2.7164E 00	1.9658E-01	-2.7310E-02	1.3816E-03	SR(OH)2	4.9563E 01	-1.9535E 01	5.6241E-01	-8.1992E-02	4.2684E-03
MGO	1.1702E 01	-5.0326E 00	2.9641E-01	-4.2811E-02	2.2023E-03	YC2	2.6685E 01	-1.4249E 01	4.4907E-01	-6.5815E-02	3.4047E-03
MGS	1.2174E 01	-3.9577E 00	3.1773E-01	-4.5792E-02	2.3491E-03	YC4	5.2624E 01	-2.6696E 01	3.5996E-01	-5.6189E-02	3.0223E-03
MGOH	2.4551E 01	-9.3818E 00	1.9666E-01	-2.7178E-02	1.3887E-03	YO	1.3514E 01	-8.7775E 00	4.0700E-01	-5.8053E-02	2.9535E-03
MG(OH)2	4.8409E 01	-1.7810E 01	1.4672E-01	-1.9833E-02	1.0312E-03	YO2	2.6764E 01	-1.6447E 01	3.9991E-01	-5.8916E-02	3.0506E-03
ALH	1.2191E 01	-3.7636E 00	2.5557E-01	-3.7261E-02	1.9406E-03	YS	1.3512E 01	-7.0421E 00	5.1174E-01	-7.3857E-02	3.7811E-03
ALC	1.2290E 01	-4.2798E 00	1.8962E-01	-2.6350E-02	1.3259E-03	ZRH	1.2751E 01	-4.7890E 00	4.1950E-01	-5.6654E-02	2.7942E-03
ALO	1.2739E 01	-5.2534E 00	1.8218E-01	-2.5793E-02	1.3185E-03	ZRC2	2.7725E 01	-1.4534E 01	3.9203E-01	-5.3832E-02	2.6810E-03
ALO2	2.5356E 01	-8.7285E 00	9.2307E-02	-1.3744E-02	7.2689E-04	ZRC4	5.3671E 01	-2.7802E 01	3.0127E-01	-4.3906E-02	2.2804E-03
AL2O	2.6158E 01	-1.2010E 01	4.2784E-01	-6.3020E-02	3.2847E-03	ZRN	1.3815E 01	-7.4335E 00	4.7870E-01	-6.5935E-02	3.2855E-03
ALS	1.2520E 01	-4.5349E 00	1.9501E-01	-2.7033E-02	1.3595E-03	ZRO	1.3296E 01	-9.0129E 00	1.9562E-01	-2.9892E-02	1.6010E-03
ALOH	2.5707E 01	-1.0624E 01	9.7901E-02	-1.1835E-02	5.8121E-04	ZRO2	2.6793E 01	-1.6151E 01	4.6988E-01	-6.4636E-02	3.2777E-03
SI2	1.2151E 01	-3.6493E 00	1.4773E-01	-2.1793E-02	1.1180E-03	ZRS	1.3694E 01	-7.4024E 00	4.6133E-01	-6.3004E-02	3.1224E-03
SIH	1.1852E 01	-3.7418E 00	1.5999E-01	-2.0629E-02	9.9897E-04	ZRF	1.3341E 01	-7.8694E 00	4.1276E-01	-5.5760E-02	2.7440E-03
SIH2	2.4664E 01	-6.8873E 00	8.3710E-02	-1.0058E-02	4.9291E-04	ZRF2	2.7367E 01	-1.5609E 01	4.2389E-01	-5.7402E-02	2.8375E-03
SIH3	3.6329E 01	-1.0556E 01	8.0945E-02	-8.6212E-03	3.9864E-04	ZRCL	1.3073E 01	-6.3789E 00	4.2411E-01	-5.7462E-02	2.8377E-03
SIH4	4.9157E 01	-1.3790E 01	-1.1493E-03	3.4469E-03	-1.9563E-04	ZRCL2	2.5230E 01	-1.2746E 01	4.1753E-01	-5.5803E-02	2.7329E-03
SIC	1.2327E 01	-5.0419E 00	1.3941E-01	-1.9363E-02	9.6202E-04	HI	1.2129E 01	-3.5797E 00	1.0764E-01	-1.3023E-02	6.1343E-04
SIC2	2.5623E 01	-1.3085E 01	-5.5227E-02	9.3363E-03	-4.9876E-04	NAI	1.1834E 01	-4.0050E 00	3.1278E-01	-4.5449E-02	2.3407E-03
SI2C	2.5549E 01	-1.1723E 01	1.7103E-01	-2.4536E-02	1.2485E-03	KI	1.1977E 01	-4.5748E 00	4.2790E-01	-6.2544E-02	3.2246E-03
SI2C2	3.8740E 01	-1.7548E 01	1.1587E-01	-1.8126E-02	9.6717E-04	RBI	1.1156E 01	-3.8605E 00	9.8534E-02	-1.2625E-02	6.0357E-04
SIN	1.2399E 01	-5.4876E 00	9.5301E-02	-1.3369E-02	6.9396E-04	CAI	1.2233E 01	-4.6155E 00	5.6168E-01	-8.2693E-02	4.2782E-03
SIO	1.3413E 01	-8.8710E 00	1.5042E-01	-1.9581E-02	9.4828E-04	CAI2	2.3330E 01	-8.0235E 00	4.6360E-01	-7.0115E-02	3.6769E-03
SIO2	2.6803E 01	-1.3431E 01	1.1642E-01	-1.6533E-02	8.5387E-04	BAH	1.2320E 01	-4.1428E 00	7.7258E-01	-1.0961E-01	5.5225E-03
SIS	1.3182E 01	-7.1147E 00	1.9300E-01	-2.5826E-02	1.2648E-03	BAO	1.3011E 01	-7.6451E 00	7.9163E-01	-1.1269E-01	5.6857E-03
P2	1.3256E 01	-6.2826E 00	4.0609E-01	-5.8731E-02	3.0089E-03	BAS	1.3090E 01	-6.5677E 00	8.1446E-01	-1.1588E-01	5.8444E-03
PH	1.2080E 01	-4.6439E 00	3.4114E-01	-4.8796E-02	2.5057E-03	BAF	1.2670E 01	-7.9603E 00	6.1272E-01	-8.5894E-02	4.2911E-03
PH2	2.3957E 01	-7.7002E 00	2.5689E-01	-3.3800E-02	1.6716E-03	BAF2	2.4712E 01	-1.4232E 01	7.7802E-01	-1.1059E-01	5.5788E-03
PH3	3.8391E 01	-1.3229E 01	1.0422E 00	-1.5646E-01	8.2658E-03	BACL	1.2932E 01	-7.3402E 00	7.7626E-01	-1.1013E-01	5.5480E-03
CP	1.2499E 01	-6.4715E 00	1.2361E-01	-1.7411E-02	8.9780E-04	BACL2	2.4320E 01	-1.2263E 01	6.8097E-01	-9.7615E-02	4.9449E-03
NP	1.3288E 01	-8.5020E 00	4.2728E-01	-6.2564E-02	3.2412E-03	BAOH	2.5317E 01	-1.2033E 01	8.1954E-01	-1.1457E-01	5.7461E-03
PO	1.2852E 01	-7.5018E 00	3.4201E-01	-4.9463E-02	2.5445E-03	BA(OH)2	4.9843E 01	-2.0406E 01	6.6675E-01	-9.3297E-02	4.7023E-03
PS	1.2366E 01	-6.5367E 00	2.7442E-01	-3.8810E-02	1.9646E-03	LAC2	2.6601E 01	-1.3780E 01	2.0185E-01	-2.4577E-02	1.1227E-03
S2	1.2960E 01	-5.0952E 00	1.8027E-01	-2.4324E-02	1.2049E-03	LAO	1.3205E 01	-9.0196E 00	1.0891E-01	-9.2430E-03	2.7882E-04
HS	1.2019E 01	-4.2922E 00	1.4913E-01	-1.8666E-02	8.9438E-04	LAO2	2.6549E 01	-1.6755E 01	1.5183E-01	-1.7590E-02	7.6542E-04
H2S	2.4632E 01	-8.4616E 00	1.7014E-01	-2.0236E-02	9.5782E-04	LAS	1.3327E 01	-6.9918E 00	2.5972E-01	-3.1783E-02	1.4512E-03
CS	1.3436E 01	-8.5574E 00	1.8754E-01	-2.5507E-02	1.2735E-03	CEC2	2.6635E 01	-1.4052E 01	2.0849E-01	-2.6096E-02	1.2216E-03
CS2	2.6653E 01	-1.2459E 01	1.3229E-01	-1.8616E-02	9.5380E-04	CEC4	5.2964E 01	-2.6907E 01	6.8877E-02	-7.5463E-03	3.1996E-04
COS	2.6743E 01	-1.4777E 01	1.1123E-01	-1.5195E-02	7.7399E-04	CEO	1.3360E 01	-9.3539E 00	2.3843E-01	-2.9163E-02	1.3343E-03
NS	1.2543E 01	-5.9563E 00	2.0901E-01	-2.8986E-02	1.4621E-03	CEO2	2.6507E 01	-1.6775E 01	1.6310E-01	-1.9846E-02	9.0402E-04
SO	1.2929E 01	-6.0100E 00	1.6253E-01	-2.1665E-02	1.0676E-03	CES	1.2888E 01	-6.9236E 00	2.6354E-01	-3.2823E-02	1.5213E-03
SH-	2.3294E 01	-8.5526E 00	6.5579E-01	-8.4089E-02	4.0414E-03	NDC2	2.6682E 01	-1.3105E 01	1.2609E-01	-1.7102E-02	8.5346E-04
HCL	1.2528E 01	-5.1827E 00	1.8117E-01	-2.4014E-02	1.1994E-03	NDC4	5.2961E 01	-2.5970E 01	-1.7939E-02	2.2173E-03	-9.3609E-05
NACL	1.1768E 01	-4.9884E 00	2.3975E-01	-3.4837E-02	1.8034E-03	NDO	1.3443E 01	-8.1399E 00	1.6252E-01	-2.1328E-02	1.0349E-03
KCL	1.2173E 01	-5.6860E 00	4.3260E-01	-6.2915E-02	3.2401E-03	NDO2	2.6535E 01	-1.6013E 01	8.7058E-02	-1.2005E-02	6.0420E-04

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1973A&A....23..411T

McBride, B.J., Heimel, S., Ehlers, J.G., Gordon, S. 1963, NASA SP–3001.
Merrill, P.W., Deutsch, A.J., Keenan, P.C. 1962, *Astrophys. J.* **136**, 21.
Morris, S., Wyller, A.A. 1967, *Astrophys. J.* **150**, 877.
Novikov, M.M., Gurvich, L.V. 1966, *Opt. Spectr.* **19**, 76.
O'Hara, D.A.G. 1970, *J. Chem. Phys.* **52**, 2992.
Peach, J.V. 1968, *Monthly Notices Roy. Astron. Soc.* **139**, 403.
Querci, M., Querci, F. 1970, *Astron. & Astrophys.* **9**, 1.
Russell, H.N. 1934, *Astrophys. J.* **79**, 281.
Sanford, R.F. 1942, *Publ. Astron. Soc. Pacific* **54**, 158.
Shenyavskaya, E.A., Maltsev, A.A., Gurvich, L.V. 1966, *Opt. Spectr.* **21**, 374.
Spinrad, H., Kaplan, L.D., Connes, P., Connes, J., Kunde, V.G., Maillard, J.P. 1970, Proceedings of the conference on late-type stars, Kitt Peak National Observatory Contribution No. 554, p. 59.
Stearns, C.A., Kohl, F.J. 1969, NASA TN D–5027.
Stearns, C.A., Kohl, F.J. 1970, NASA TN D–5653.
Stearns, C.A., Kohl, F.J. 1971, NASA TM X–67844.
Suess, H.E., Urey, H.C. 1956, *Rev. Mod. Phys.* **28**, 53.
Swings, P. 1958, Handbuch der Physik, Ed. S. Flügge **50**, 109, Springer, Berlin · Göttingen · Heidelberg.
Tomley, L.J., Wallerstein, G., Wolff, S.C. 1970, *Astron. & Astrophys.* **9**, 380.
Tsuji, T. 1964, *Ann. Tokyo Astron. Obs.*, 2nd ser. **9**, 1.
Tsuji, T. 1966, *Publ. Astron. Soc. Japan* **18**, 127.
Tsuji, T. 1972, To be published.
Vardya, M.S. 1966a, *Monthly Notices Roy. Astron. Soc.* **134**, 347.
Vardya, M.S. 1966b, *Observatory* **86**, 32.
Vardya, M.S. 1967, *Mem. Roy. Astron. Soc.* **71**, 249.
Wahlbeck, P.G., Gilles, P.W. 1967, *J. Chem. Phys.* **46**, 2465.
Warner, B. 1968, *Monthly Notices Roy. Astron. Soc.* **138**, 229.
White, W.B., Johnson, S.M., Dantzig, G.B. 1958, *J. Chem. Phys.* **28**, 751.
Wiehr, E., Stellmacher, G., Schröter, E.H. 1968, *Astrophys. Letters* **1**, 181.
Zambov, K.F., Margrave, J.L. 1966, *J. Chem. Phys.* **45**, 3167.
Zambov, K.F., Margrave, J.L. 1967, *J. Phys. Chem.* **71**, 2893.

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